

CS5185: MULTIMEDIA TECHNOLOGIES AND APPLICATIONS

Effective Term

Semester A 2025/26

Part I Course Overview

Course Title

Multimedia Technologies and Applications

Subject Code

CS - Computer Science

Course Number

5185

Academic Unit

Computer Science (CS)

College/School

College of Computing (CC)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Nil

Precursors

Nil

Equivalent Courses

Nil

Exclusive Courses

Nil.

Part II Course Details

Abstract

The course aims at providing students with theoretical and technical understanding on multimedia components communications, and systems. The course covers contemporary, interactive multimedia technology and communication systems, focusing on types, applications, and theories of operation. Basic technologies such as multimedia data representation, compression, retrieval and communication will be covered in an integrated manner. On completion of the course, students should be able to understand the fundamental concepts and make critique to the technologies associated with various multimedia data types such as image, video, audio, graphics and animation.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Explain approaches to represent multimedia data in digital format and identify their properties based on the human visual and auditory perception.	20	x		
2	Discuss image, video and audio properties in the frequency domain to identify important components to be encoded.	20		x	
3	Explain the major steps in some of the image, video and audio compression standards, including quantization, coding techniques and coding standards.	25		x	
4	Explain multimedia streaming and QoS for real-time multimedia data transmission.	15	x		
5	Apply lossless and lossy compression techniques on multimedia data.	20		x	x

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Lecture	Student will engage in formal lectures to learn basic technologies such as multimedia data representation, frequency domain features, human perception, lossy and lossless compression, compression standards, etc..	1, 2, 3, 4, 5	2 hours/ week

2	Tutorial	Students will participate in some class exercises each week during the tutorial sessions. In particular, they will engage in group discussions on how to solve various multimedia-related problems.	1, 2, 3, 4, 5	1 hour/ week
3	Project	Students will engage in solving scenarios-based problems that involve applying related techniques learnt from the lectures. While the problem is being solved, the students will also engage in discovering the rationale behind the particular approach and explaining their proposed solutions.	2, 5	

Assessment Tasks / Activities (ATs)

ATs		CILO No.	Weighting (%)	Remarks ("-" for nil entry)	Allow Use of GenAI?
1	Quiz	1, 2, 4	20	-	No
2	Course Project	2, 5	20	-	Yes

Continuous Assessment (%)

40

Examination (%)

60

Examination Duration (Hours)

2

Minimum Examination Passing Requirement (%)

30

Additional Information for ATs

For a student to pass the course, at least 30% of the maximum mark for the examination must be obtained.

Assessment Rubrics (AR)**Assessment Task**

Quiz (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Capacity in understanding the key concerns of multimedia data and techniques

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Note even reaching marginal levels

Assessment Task

Course Project (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to apply multimedia techniques on a practical application

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Note even reaching marginal levels

Assessment Task

Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to analyse and evaluate multimedia data and techniques and apply multimedia techniques on applications

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Note even reaching marginal levels

Assessment Task

Quiz (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Capacity in understanding the key concerns of multimedia data and techniques

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate

Failure

(F) Note even reaching marginal levels

Assessment Task

Course Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to apply multimedia techniques on a practical application

Excellent

(A+, A, A-) High

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Assessment Task

Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to analyse and evaluate multimedia data and techniques and apply multimedia techniques on applications

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Part III Other Information

Keyword Syllabus

Multimedia Data Compression, Multimedia Data Representation, Image and Video Compression, Digital Audio, Multimedia Database Systems.

Syllabus:

- Image Representation
- Color Science and Color Models
- Lossless and Lossy Compression
- JPEG Image Compression Standard
- Video Representation
- Basic Video Compression Techniques
- Video Coding Standards: H.26X and MPEG
- Basics of Digital Audio
- Audio Compression

Reading List**Compulsory Readings**

Title	
1	Nil

Additional Readings

Title	
1	Ze-Nian Li and Mark Drew (2004). Fundamentals of Multimedia. Prentice Hall.
2	J. Buford (1994). Multimedia Systems. Addison Wesley.
3	Jens-Rainer Ohm (2015). Multimedia Signal Coding and Transmission. Springer.
4	Mario Marques da Silva (2012). Multimedia Communications and Networking. CRC Press.